

Chapter – 6

Conclusion

This IoT architecture for battery charging management not only meets the immediate challenges posed by the growing demand for EV infrastructure but also promotes sustainable practices through effective utilization of renewable energy sources. Future developments will focus on refining this architecture and expanding its capabilities to further enhance energy management in electric vehicle applications. By leveraging real-time data analytics and centralized monitoring, this project positions itself as a forward-thinking solution in the rapidly evolving landscape of electric vehicle infrastructure, ultimately contributing to a greener transportation ecosystem.

Future Scope

The future scope of this project is vast and promising. It includes integrating Vehicle-to-Grid (V2G) technology to enable bidirectional power flow, incorporating wireless charging solutions for enhanced convenience, and developing megawatt-scale chargers for heavy-duty vehicles. Expanding public charging infrastructure globally and integrating battery-swapping technologies are also key areas of focus. Additionally, leveraging IoT for real-time monitoring and predictive maintenance will ensure operational efficiency. These advancements will contribute significantly to achieving global decarbonization goals by supporting the electrification of transportation sectors.